Luck, Skill, and Trader Heterogeneity in Prediction Markets:

Evidence from Polymarket's Crypto Markets

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August 2026

Project AI Usage and Reflections (Please insert this after the cover page)

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Claude, ChatGPT. I used AI for coding and to improve the fluency and naturalness of my English writing.

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AI coding and language polishing improve my efficiency, but the work itself would not change. I used Claude to generate bootstrapping code, but the code was wrong in details. I found it using ChatGPT to cross-check the code. I used Claude to generating clustering feature calculation code, and I also used it to generate test, then the test pointed out the wrong calculation. The AI use ultimately improved my work mainly by improving efficiency.

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academic integrity: When AI assisted me with coding and language polishing, I used it to implement the methodology I designed, instead of asking it to generate the research approach or conclusions. The research questions, methodology, and other analysis were developed independently by me.
accuracy/verification: I used both ChatGPT and Claude to cross-check codes.
clear authorship: I only used AI to do coding and language polishing, so clearly the content was my own work and authorship.

Section 4: Additional information you may wish to include

Declaration: I acknowledge the use of [Claude & ChatGPT]. I confirm that no content generated by AI has been presented as my own work **(please do not add your name or CID here and only the date):**
Date……………09 August 2026………………………………………

Abstract

This paper asks whether traders in Polymarket's cryptocurrency betting markets earn returns that luck alone cannot explain, and whether skill is concentrated in particular kinds of trader. We reconstruct minute-level profit and loss for 653,910 wallets from the complete on-chain record of 218.9 million order-book fills between May 2025 and February 2026, and treat each wallet as an actively managed portfolio with its Bitcoin exposure removed. Wallets are clustered on 18 purely behavioral features and the partition isolates a high-frequency group that accounts for 10.3% of wallets and a residual group of directional traders. Skill is then tested within each group against three bootstrap procedures drawn from the mutual-fund literature, whose size and power are calibrated on the panel itself. The empirical finding shows that skill exists before clustering. However, it is not located in high-frequency traders. Positive alpha is attributed to 17.6% of directional traders but only 3.8% of high-frequency wallets, whose median t-statistic is −0.81 and who lose systematically.

Introduction

Prediction markets, a concept that has been studied for decades, have experienced rapid growth in recent years, mainly due to advances in blockchain technology, regulatory changes, and U.S. presidential elections. Among all prediction market platforms, Polymarket is the largest (Polymarket, 2026). Since August 2025, Polymarket's total value locked (TVL) has been growing steadily (DefiLlama, 2026), as daily notional trading volume reached as high as around $ 0.48 billion (Blockworks, 2026).

All Polymarket transactions are recorded on-chain, including every action by every wallet. In traditional financial markets these data are inaccessible to researchers, making Polymarket a valuable dataset for studying trader behavior and market microstructure. As a result, a growing number of working papers have emerged in recent months, examining Polymarket from a variety of perspectives, including systematic pricing models (Dalen, 2026), market manipulation, regulatory concerns (Sirolly et al., 2026), and the performance of market participants (Reichenbach & Walther, 2025; Akey et al., 2026).

This study asks whether traders in Polymarket's crypto related betting earn returns that luck alone cannot explain. And whether skill, if present, is concentrated in some kinds of trader.

Directional traders on centralized crypto exchanges are known to obtain an edge through model-driven strategies, exhibiting both timing and asset-selection skill (Conlon, De Mingo-López & Urquhart, 2025). Our initial hypothesis was that in Polymarket markets tied directly to cryptocurrency price dynamics, directional traders would exhibit no such skill, and that this absence may follow from the platform's on-chain transparency or polymarket market design. The skill hypothesis is tested and rejected. Directional traders do earn alpha in these markets. Instead the high-frequency market makers and arbitragers lose systematically.

The empirical design treats each wallet as an actively managed portfolio. This paper uses three bootstrap procedures drawn from the mutual fund literature: the residual resampling of Kosowski et al. (2006), the joint calendar resampling of Fama and French (2010), and the threshold variant of Harvey and Liu (2022). We transfer the methods developed for mutual funds to a sparse, severely unbalanced panel of on-chain wallets. Wallets are clustered on trading behaviors before any performance measure is examined, separating directional traders from the high-frequency wallets.

The analysis yields findings contrary to expectations. First, skill is present in the full sample. At the 95th percentile of the cross-section the actual t-statistic is 3.82 against a null distribution of 2.06. A false discovery rate decomposition attributes genuinely positive alpha to 15.4% of wallets. Second, skill is not located where microstructure theory would place it. Among high-frequency wallets, genuinely positive alpha is attributed to 3.8% of them. They lose systematically. However, among the directional traders 65.2% earn positive statistics and positive alpha is attributed to 17.6%.

The paper makes three contributions. First, it brings the fund-evaluation standard to an on-chain prediction market. Existing work on prediction market relies almost entirely on accuracy measures inherited from the gambling literature. Second, it studies a segment whose economics differ from the rest of the venue and considering trader types. This segment has attracted little attention relative to the election markets. Third, the findings contribute to market design in the future. Prediction markets have attracted attention as a promising direction for development (CBOE, 2026), and Polymarket continues to launch new cryptocurrency events and adjust its mechanisms. Therefore, insights into trader behavior and profits are valuable to both regulators and platform.

Section 2 reviews the relevant literature. Section 3 describes the data and the construction of wallet-level PnL. Section 4 develops wallet classification. Section 5 sets out the testing methodology, Section 6 reports the results, and Section 7 concludes. Appendix 1 has the Github link where we put the code in.

2 Related Literature

2.1 Separating Luck from Skill: KTWW, Fama-French, Harvey-Liu

Skills versus luck is one of the most important topics in financial markets. Traditional finance has provided a set of methods to analyze funds' performance. Distribution after bootstrapping is a golden rule. Kosowski et al. (2006) develop the KTWW bootstrap. Fama and French (2010) extend this by bootstrapping the entire cross-section simultaneously. Harvey and Liu (2022) further propose an alternative bootstrap approach that addresses the limitations of previous methodologies regarding test size and power. An alternative perspective defines skill using performance persistence (Maroof, Javid & Badshah, 2017). False discovery rate (FDR) methods also provide complementary tools (Barras, Scaillet & Wermers, 2010).

Most of crypto betting events in Polymarket can be viewed as cryptocurrency derivatives, thus the literature on skilled traders in cryptocurrency funds provides a useful point for comparison. Evidence shows that cryptocurrency funds, as sophisticated investors, generate alpha, exhibit performance persistence, and demonstrate market timing ability (Conlon, De Mingo-López & Urquhart, 2025).

2.2 Trader Heterogeneity and Classification

We know Polymarket has diverse trader types. Arbitrage trading is prevalent on Polymarket (Saguillo et al., 2025). Wash trading also takes up a large volume (Sirolly et al., 2026). Thus, trader classification is essential for conducting more detailed studies.

Rather than conceptual categorization, trader classification is usually more data-driven in literatures. Kirilenko et al. (2017) classify traders into High-Frequency Traders, Market Makers, Small Traders, Fundamental Buyers, Fundamental Sellers, and Opportunistic Traders using three criteria including volume, end-of-day position ratio, and intraday inventory pattern. Mitts and Ofir (2026) develop a composite score computed at the wallet & market pair level as a weighted sum of five signals, where higher scores generate higher extra profits. Using clustering algorithms, Benford's law, and distribution tests, researchers can identify potential wash traders (Cong L et al., 2023; Sirolly et al., 2026).

2.3 The Prediction Market Literature

Prediction markets have been recognized for their effective information aggregation function for a long time (Arrow et al., 2008). The emergence of Polymarket and Kalshi in recent years has provided new data sources for academic research. The majority of studies focus on the political betting. For example, some papers analyze trading data from Polymarket during the 2024 U.S. presidential election, examining arbitrage opportunities, price discovery, and information transmission (Tsang & Yang, 2026; Ng H et al., 2026). Other papers mainly focus on systematic pricing models (Dalen, 2026), market manipulation and regulatory concerns (Sirolly et al., 2026).

Because modern prediction markets are still relatively new, the literature on skilled traders and trader performance on Polymarket remains limited, with the following 2 working papers being the most related. Akey et al. (2026) use Polymarket's on-chain data and discover extreme profit concentration, well-calibrated prices, and genuine skilled traders using excess hit rates. It identifies taker activity as the key predictors for trading loss. Reichenbach and Walther (2025) offer an explorative study of trader skill on Polymarket. The authors confirm the existence of trader skill through the distribution of profits and losses, skill versus luck test, and the persistence of returns over time, then further identify some systematic differences between winners and losers.

2.4 Research Gaps

Reichenbach and Walther (2025) and Akey et al. (2026) represent the key related literature to this study. Akey et al. (2026) leverages Polymarket's on-chain data to show existence of genuine skilled traders using excess hit rates. Reichenbach and Walther (2025) tests the existence of Polymarket skilled traders through PnL distributions, skill-versus-luck tests, and return persistence over time. However, there are several critical gaps.

First, prior work treats Polymarket as homogeneous despite hosting heterogeneous markets. Crypto betting and other betting events differ fundamentally. Political and weather contracts are primarily traded on Polymarket and represent aggregation of private information and price discovery. Crypto contracts, however, widely trade on CEXs and DEXs with superior liquidity and lower fees. Plus, crypto markets have less private information. This suggests Polymarket's crypto section may simply track external prices rather than discover them independently. Directional traders with sophisticated ML models may not have incentive to trade crypto on Polymarket over centralized exchanges.

Second, prior work does not distinguish between trader types. Mixing different trader behaviors obscures Polymarket's true mechanisms. An arbitrageur and a directional trader earn returns through opposite logics. Pooling the two yields an average can mask offsetting patterns entirely.

At its core, this study asks whether skilled traders exist in Polymarket's crypto-related betting markets, and if so, which kind of trader they are. It uses a behavioral classification estimated without reference to performance, and three bootstrap tests drawn from the mutual fund literature.

3 Data and Return Construction

3.1 Data Structure

On Polymarket, an event is a topic, such as the path of the Bitcoin price in January 2026. Under each event there are one or more markets, each corresponding to a specific tradeable proposition. For example, whether BTC closes above $100,000 on 31 January. The crypto segment of Polymarket studied here is dominated by events that related to different price thresholds and horizons of some specific coins, such as BTC go up or down in 1 hour.

Every market is a binary contract identified by a condition_id. It is backed by two outcome tokens, YES and NO, each with its own asset identifier. One unit of USDC can be split into one YES and one NO token, and the pair can be merged back. At resolution the token matching the realized outcome redeems at $1 and the other at $0. Between a market's opening and resolution, the outcome tokens trade freely on a central limit order book, matched in the same way as on a conventional centralized exchange. Settlement of every match is written to the blockchain. Full details are available in Polymarket's official documentation.

Using Polymarket Gamma API, we obtain Polymarket events and markets metadata. The specific filtering procedure is documented in Appendix 2. Then we retrieve the complete on-chain OrderFilled event history via Goldsky Turbo Pipelines. The full schema is documented in Appendix 3. We restrict our analysis to markets that open and end within 2025-05-01 and 2026-02-05. We have total orderfilled records 218,863,487 with Block range: 70,987,253 to 82,564,403, covering 148,131 Markets and 121,498 Events.

Through extensive checks, we confirm that no material trading activity is missing in the final dataset. The structure of the raw records is important. Each economic trade generates two classes of OrderFilled event: one taker row, whose counterparty field is the Exchange contract address, and one or more maker rows, whose counterparty is the taker's wallet.

The value recorded in the fee field cannot be taken as the true value. Inspection of the on-chain records shows that a fee undergoes several further transfers after the fill is recorded, so this field indicates only that a fee occurs, not the exact amount. Furthermore, fees were introduced late in the platform's history and the schedule has changed repeatedly since.

We reconstruct the matching and fee mechanics of Polymarket V1 by cross-referencing the official changelog against the smart-contract source code published by Polymarket on GitHub. This recovers the usage of the CTF Exchange and NegRisk CTF Exchange addresses, the trade-level event structure, the taker-maker counterparty convention, and the settlement-stage fee mechanics.

3.2 Reconstructing Wallet-Level Profit and Loss

For each wallet-token pair we construct a chronologically event stream that merges three sources: order fill, price, and settlement.

1. OrderFill events. All trades executed by this wallet for this token.
2. Price events. The mean execution price of the token in each block in which it traded anywhere in the market, by aggregating all fills across all wallets.
3. Settlement events. A terminal event per token, placed at (last observed block $+1$), with the outcome price 0 or 1.

Within a block, events are processed in the order of trading price, settlement, and then fill (fills ordered by log_index).

Polymarket trading fees involve the exchange contract and some fee modules, which are not directly recoverable from on-chain indexers. The fee field in our Goldsky data is not an accurate number. On-chain records show that an orderfill is followed by a further sequence of transfers, so the fee ultimately paid differs from the value in our Goldsky data fee field. Also, the maker fee is zero, plus some extra Maker Rebates or Liquidity Rewards.

After cross-checking the Github Polymarket contract source code and on-chain data, the net taker fee rate can be approximated as

$$f(p) = 0.25\,[p\,(1-p)]^2,$$

which is bounded above by $f(0.5)$. In the reconstruction, a non-zero fee is applied if and only if the row is a taker row and the on-chain fee field is strictly positive. The fee is on different legs depending on trading direction. On a purchase the fee is deducted from the share leg. On a sale it is deducted from the USDC leg.

Our engine maintains two parallel ledgers for every wallet, the ledger with fees and the no-fee ledger, just like the net/gross analysis in Kosowski et al. (2006) and Fama and French (2010).

Gross exposure E_t at time t is the absolute value of all unsettled positions,

$$E_t = \sum_{a\,:\,\text{unsettled}} \left| pos_{a,t}\, m_{a,t} \right|$$

At the settlement event the token is marked to $\pi \in \{0,1\}$, so that the realised gain or loss on a position held to expire enters PnL at that time. From that moment the token is excluded from the sum exposure, and its mark is frozen against subsequent events. The resulting series are aggregated to minute-level end-of-interval states. States are recorded only in minutes containing at least one event, yielding a sparse representation. Then cumulative profit and loss is forward-filled onto a common calendar and differenced to obtain daily profit and loss:

$$\mathrm{PnL}_{i,d}^{\mathrm{daily}} = \mathrm{PnL}_{i,d} - \mathrm{PnL}_{i,d-1}$$

Forward-filling ensures that overnight revaluations and settlement jumps are attributed to the correct day, and those days on which no event occurs contribute exactly zero.

When calculating returns, unlike a mutual fund, a wallet has no net asset value, so a denominator must be constructed. We use capital at risk, defined as

$$C_{i,d} = \max\left\{\max_{t \in d} E_{i,t}, \quad E_{i,\tau(d)}\right\}$$

where the first term is the intraday peak exposure and the second is the end-of-day exposure carried forward from the most recent day $\tau(d)$ on which the wallet had an event. This avoids a degenerate denominator for intraday traders who close all positions before the day ends. The carried-forward term ensures that a wallet holding a position through a quiet day is still recorded.

The daily return is defined only when capital at risk exceeds a floor, to avoid the errors from the fee approximation:

$$r_{i,d} = \begin{cases} \dfrac{\mathrm{PnL}_{i,d}^{\mathrm{daily}}}{C_{i,d}}, & C_{i,d} > \phi_i, \\ \text{missing}, & \text{otherwise,} \end{cases} \qquad \phi_i = \max\left\{0.50, \ \rho \cdot \max_t E_{i,t}\right\}.$$

The relative component of the floor is set at $\rho = 2\%$, so that days on which the recorded exposure consists only of fee residue are classified as non-observations. A wallet holding no position has zero capital at risk and contributes no observation. A wallet that opens and closes within the day is evaluated against its intraday peak. This reconstruction was validated by randomized scenarios, synthetic wallets, and randomly sampled wallets.

4 Clustering Traders

The motivation for clustering is that trading behavior in this market is heterogeneous. Market makers and arbitrageurs, whose edge derives from speed, may plausibly be skilled. Directional traders, for some reasons, either an unwillingness to reveal private abilities in a fully transparent venue, or the absence of such ability at all, may not have skills. Clustering is thus the pivotal step in our analysis. We partition the 653,910 wallets using 18 behavioral features.

4.1 Notations and Adjustments

We use the complete on-chain OrderFilled event history data as we said previously. All 653,910 wallets are retained. No activity filter is imposed.

Notation is as follows. Wallet u; fill i with notional u_i USDC, share quantity q_i, and price p_i; market c (condition_id); outcome token a (asset).

As said before, we use the fee approximation. Thus, a position is treated as flat when it falls below a threshold, to clean the dust the fee leaves:

$$\text{flat}_{u,a}(t) \Leftrightarrow |\text{pos}_{u,a}(t)| \ \le \ \max\left(\frac{\$0.50}{\bar{p}_{u,a}},\ \rho \cdot \text{peak}_{u,a}\right)$$

where $\text{peak}_{u,a} = \max_t |\text{pos}_{u,a}(t)|$ and $\bar{p}_{u,a}$ is the average trade price in that token. We set $\rho = 2\%$.

Furthermore, positions are not zeroed at market resolution, because redemption generates no OrderFilled event. We truncate at the last block in which that token traded. This is a proxy for the resolution time.

For each (wallet, token) pair, fills are ordered by timestamp. A holding period opens at the fill that moves the position from flat to non-flat and closes at the first subsequent fill that returns it to flat. holding periods that never close are treated as held to settlement.

4.2 Feature design

The 18 features are purely behavioral. No performance variable, such as realised PnL or win rate, enters the feature set. The restriction follows Kirilenko et al. (2017), and Mitts and Ofir (2026).

Scale, breadth, timing features (8 features)

Feature	Definition	Meaning
log_n_fills	$\ln(1 + n_{\text{fills}})$	Overall trading activity
log_volume_usdc	$\ln(1 + \sum_i u_i)$	Capital deployed
log_avg_fill_usdc	$\ln(1 + \sum_i u_i / n_{\text{fills}})$	Typical size per fill
log_n_markets	$\ln(1 + \|\{c\}\|)$	Breadth of market coverage
log_fills_per_market	$\ln(1 + n_{\text{fills}}/\|\{c\}\|)$	Repeated trading or one-off entry
log_fills_per_active_day	$\ln(1 + n_{\text{fills}}/n_{\text{active days}})$	Intensity on days the wallet is active
active_minute_density	$\min(1,\ n_{\text{active minutes}}/\max(\text{span}_{\text{minutes}},\ 1440))$	Persistence of presence or proxy for automation
burst_5plus_frac	share of active minutes containing five or more fills	Sub-minute order clustering or machine-speed execution

Microstructure features

Feature	Definition	Meaning
maker_frac	fraction of fills executed as the resting side	Liquidity provision
micro_fill_frac	fraction of fills with notional below $1	Economically negligible orders
int_round_frac	fraction of fills with integer-rounded share quantity or notional	Manual vs programmatic order entry

Price positioning features

Feature	Definition	Meaning
vw_avg_price_yes	$\text{mean}_i\left(p_i^{\text{yes}}\right)$	Where on the probability axis the wallet trades
vw_std_price_yes	$\text{sd}_i\left(p_i^{\text{yes}}\right)$	Breadth of the probability range traded
intra_market_price_std_mean	$\text{mean}_c\left[\text{sd}_{i\in c}\left(p_i^{\text{yes}}\right)\right]$	Trading around a moving price within one market

Holding behavior features

Feature	Definition	Meaning
roundtrip_frac	closed spells / total spells, aggregated across all tokens	does the wallet exit its positions
log_median_hold_minutes	$\ln\left(1 + \text{median}_{\text{spells}}(\text{hold\_seconds})/60\right)$	Typical holding horizon, truncated at settlement
hedged_exposure_frac	see below	Share of exposure-time offset by the opposite leg

Let y and n denote the signed net positions in the YES and NO legs of a market at a given instant, after dust filtering.

$$\text{net directional exposure} = |y - n|, \qquad \text{gross exposure} = |y| + |n|$$

$$\text{hedged exposure} = |y| + |n| - |y - n|$$

Positions are step functions of time, so the quantity is integrated exactly by a sweep over fills:

$$\text{hedged\_exposure\_frac}_u = \frac{\sum_c \sum_k (|y_k| + |n_k| - |y_k - n_k|)\, \Delta t_k}{\sum_c \sum_k (|y_k| + |n_k|)\, \Delta t_k}$$

where k indexes the inter-fill intervals within market c and the final interval is truncated at settle.

Counterparty concentration feature:

$$\text{cp\_hhi}_u = \sum_j \left(\frac{s_j}{\sum_j s_j} \right)^2$$

Meaning: the degree to which a wallet transacts repeatedly against the same addresses, a standard wash-trading signal.

4.3 Clustering Procedure

Entries in the 653,910 × 18 feature matrix are median-imputed, each feature is mapped to a standard normal by a quantile transformation (n_quantiles = 1000), and the result is reduced by PCA to seven components retaining 89.0% of total variance. K-means is then run on these components (n_init = 20, max_iter = 500, seed = 42).

Because the silhouette is nearly flat over K = 4 to 12, it cannot on its own pin down K. We therefore also appeal to the prior literature regarding trader classifications, such as trader types in Kirilenko et al. (2017). Furthermore, moving from K = 7 to K = 8, key clusters are essentially unchanged. The resulting market-making cluster is stable beyond K = 8, higher K subdividing only the directional clusters. We therefore set K = 8.

This choice is not critical. Clusters are subsequently pooled into a market-making group and a directional group. The aim is the sharpest defensible boundary between trading styles. Because behavior lies on a continuum, moderate silhouette values reflect the structure of the data rather than a failed partition.

4.4 Stability and diagnostics

Silhouette (K = 8)	0.346
PCA variance retained	89.0%
Bootstrap Jaccard	0.847
Largest / smallest cluster	17.8% / 7.5%

4.5 Clustering Result

The clustering result shows 8 behavioral types. Type 6 exhibits the complete market-making signature. It comprises 10.3% of wallets, 81.2% of notional volume and 96.1% of fills.

		Size		Activity		Strategy signature					
k	**Type Description**	**% wal.**	**Med. vol.**	**Med. fills**	**Med. mkts**	**Med. hold**	**Maker**	**R-trip**	**Hedged>0**	**Burst>0**	**cp-HHI**
6	Automated MM / HFT	10.3	$2,501	166	47	14 m	0.48	0.32	75.8	73.5	0.048
4	Small-scale active	12.7	$76	16	7	52 m	0.00	0.49	16.7	4.4	0.162
2	Limit-order builders	15.2	$416	8	6	298 m	0.67	0.013	1.9	5.7	0.315
0	Short-horizon direct.	17.8	$135	4	2	277 m	0.00	1.00	2.1	1.2	0.356
7	Low-freq. long-shot	16.8	$52	4	4	2,523 m	0.00	0.00	0.3	0.8	0.343
5	Low-freq. even-odds	9.8	$34	4	3	805 m	0.00	0.00	1.0	2.6	0.371
1	One-shot, even odds	10.0	$8	1	1	558 m	0.00	0.00	0.0	0.0	1.000
3	One-shot, long shot	7.5	$7	1	1	2,008 m	0.00	0.00	0.0	0.0	1.000

k is the KMeans cluster index. All values are medians except Hedged > 0 and Burst > 0, which report the percentage of wallets in the type with a strictly positive value. Med. vol. is median notional volume per wallet; Med. hold is median position-spell duration in minutes; R-trip is the round-trip rate; cp-HHI is the HHI index of counterparty concentration.

5 Methodology for Testing Skill

5.1 Baseline Regression

We treat each wallet as an actively managed portfolio and ask whether risk-adjusted performance contains more skill than luck. The benchmark model regresses wallet returns on the single systematic factor, the return on BTC.

$$r_{i,t} = \alpha_i + \beta_i F_t + \varepsilon_{i,t}, \qquad t = 1, \dots, T_i.$$

Estimation is by OLS over the days on which wallet i has a return. Inference is based on the t-statistic and conventional standard errors are used.

There are numerical guards applied at the estimation stage. Numerically degenerate regressions are discarded, when regressions have residual variance falls to machine precision. The same guards apply to actual and bootstrapped samples. Wallets must have at least non-zero return of 8 days.

The null hypothesis to all 3 methods is that no wallet possesses real skill, $\alpha_i = 0$ for all i. The following 3 procedures differ in how the null world is constructed.

5.2 Method 1: KTWW

The KTWW procedure from Kosowski et al. (2006) resamples residuals within each wallet, while keeping the factor series fixed. Wallets are required to have $T_i \geq 60$ observations as in the original study.

For each bootstrap iteration $b = 1, \dots, B$ and each wallet i, a sequence of indices $\{s_t\}$ is drawn with replacement from $\{1,\dots,T_i\}$ and a pseudo-return series is constructed as

$$r_{i,t}^{b} = \hat{\beta}_i F_t + \hat{\varepsilon}_{i,s_t}, \qquad t = 1, \dots, T_i.$$

In this process, the intercept, which is the original alpha estimation, is omitted, so the null is imposed by construction. The factor is kept in its original chronological order, only residuals are reshuffled. Each wallet is resampled independently, and the factor series stays the same across all iterations. The regression is then estimated on the pseudo-series, retaining the intercept. Then $t^b(\hat{\alpha}_i)$ is recorded.

Because wallets are resampled independently, this design removes the cross-sectional dependence present in the data. Residual co-movement not absorbed by the benchmark model induces positive correlation among the estimated alphas, which widens the sampling distribution of extreme order statistics. So the test would over-reject.

5.3 Method 2: Fama and French

The FF procedure from Fama and French (2010) resamples calendar dates jointly across the entire cross-section. Wallets are required to have $T_i \geq 8$ observations.

Each wallet's returns are first demeaned by its own estimated alpha,

$$\tilde{r}_{i,t} = r_{i,t} - \hat{\alpha}_i,$$

producing a population in which every wallet has an in-sample alpha of exactly zero. Note that $\hat{\alpha}_i$ is estimated over the wallet's full history and subtracted uniformly; the demeaned panel supplies the resampling population, while the actual cross-section of $t(\hat{\alpha}_i)$ against which it is compared is computed from the original, undemeaned data.

A simulation run proceeds as follows. Let D denote the number of calendar days spanned by the factor series. Draw D days with replacement from $\{1, \dots, D\}$, this draw is shared by every wallet and by the factor. For each wallet, retain those drawn days on which it has an observation, in this case with a day drawn c times contributing c identical rows. Returns and factors are therefore never decoupled.

Wallets for which the number of distinct retained days falls below 8 are dropped from that run. Then the regression, with intercept, is estimated again using OLS on the stacked pseudo-sample and $t^b(\hat{\alpha}_i)$ is recorded.

FF's method preserves the cross-sectional correlation among wallet returns and the joint distribution of returns and factor. The cost is that each wallet's effective sample size becomes random, resulting in the null distribution too fat in the tail.

5.4 Method 3: Harvey and Liu Thresholding

The Harvey–Liu procedure from Harvey and Liu (2022) retains the joint resampling scheme of FF and adds a filter that removes wrong t-statistics due to sample-size distortion. Wallets are required to have $T_i \geq 12$ observations.

Stage one is to calculate complete-sample bands. For each wallet, working with the zero-alpha returns $\tilde{r}_{i,t}$ and restricted to that wallet's own observation dates, draw T_i indices with replacement and re-estimate the regression on the paired sample. The sample size is held fixed at T_i, which isolates the sampling variation of the t-statistic under a correctly sized sample. Repeating this 1000 times would yield a bootstrap distribution of t-statistics, and the band is defined using quartiles $\hat{q}_{25,i}$ and $\hat{q}_{75,i}$.

$$\mathcal{B}_i = \left[\hat{q}_{25,i} - c \cdot \mathrm{IQR}_i,\ \ \hat{q}_{75,i} + c \cdot \mathrm{IQR}_i\right], \qquad \mathrm{IQR}_i = \hat{q}_{75,i} - \hat{q}_{25,i}.$$

This functional form is a fence rule, with a tuning parameter constant c.

Stage two is the threshold resampling. The FF resampling is executed exactly as before. But in each iteration, a wallet's realized t_i^b falls outside Bi is excluded from that iteration. Exclusion is per iteration and not permanent.

The band is applied to the null distribution instead of the actual data. In the actual data, every wallet is estimated on its complete history. In this way, extreme t-statistics in the actual cross-section arise only from luck or from skill.

5.5 Inference

For each iteration b, all three methods give the cross-section of bootstrapped t-statistics, and we record percentiles $\hat{P}_b$ and summary statistics. For a statistic S with actual value S^{act}, the right-tail p-value is

$$p_{\text{right}}(S) = \frac{1}{B}\sum_{b=1}^{B} \mathbf{1}\{S_b \geq S^{\text{act}}\}$$

$$\text{Correction } p_{\text{right}}(S) = \frac{\sum_{b=1}^{B} \mathbf{1}\{S_b \geq S^{\text{act}}\} + 1}{B+1}$$

with the left-tail p-value defined symmetrically. We report B = 1,000 iterations. The primary statistic is the 95th percentile of the cross-sectional $t(\hat{\alpha})$ distribution, designated in advance, with the 90th and 99th percentiles reported alongside. The 99th percentile statistic is not correctly sized in this panel under any configuration and is therefore reported descriptively only. The size and power properties of the individual statistics are established directly by the calibration exercise described next. We additionally report a false discovery rate decomposition.

5.6 Calibrating the Thresholding

Harvey and Liu (2022) report that threshold c = 2.0 as calibration settings in their monthly mutual fund data, but state that no bootstrap procedure is universally optimal. Because our daily wallet panel here is far sparser and more unbalanced, c needs to be recalibrated. The following calibration procedure follows Harvey and Liu (2022).

The calibration builds synthetic world in which the answer is known.

Residuals are drawn with replacement from each wallet's own history. A single calendar map σ is drawn with replacement and shared across all wallets. σ is a shared instruction sheet, for example, for day 1 use day 3, for day 2 use day 5. for each of its own observation dates d, wallet i takes the residual it recorded on date $\sigma(d)$, or, if it did not trade that day, an independent draw from its own residuals.

Sharing σ aligns wallets on common dates, so that any pair trading on the mapped date receives contemporaneous residuals and the co-movement in the data survives into the simulation. Iterating over the wallet's own dates fixes the sample size at T_i by construction, so the sampling distortion the test is designed to detect is not manufactured by the simulation.

The calibration runs under two configurations.

In the first configuration, no wallet is given any skill: $\alpha_i = 0$ for every wallet, so anything the test flags is a false positive. We build 100 such panels, run the test on each, and count how often it rejects. If the test is working properly, that count should come out near the 10% we set as the nominal level.

In the second configuration, a randomly chosen 5% of wallets are given a genuine positive alpha, sized to an information ratio of 0.75 and converted to a daily figure. Here rejections are the right answer, and the share of panels on which the test rejects is its power.

On each synthetic panel we run the full two-stage Harvey–Liu procedure, on our own data, the calibration settles on $c = 4.0$.

6 Results and Interpretation

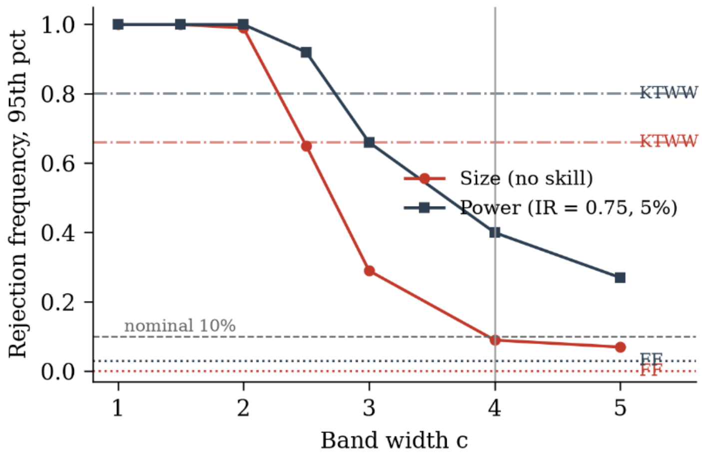
6.1 Estimation Sample

Among the 653,910 wallets in the rebuilt PnL data, 218,959 wallets with $T \geq 8$ enter the Fama–French (FF) test, 176,898 with $T \geq 12$ enter the Harvey–Liu (HL) test, and 16,027 with $T \geq 60$ enter the Kosowski (KTWW) test.

6.2 Calibration of the Bootstrap Procedures

Following Harvey and Liu (2022), synthetic panels are constructed with the size configuration and the power configuration. Each configuration is replicated 100 times and the entire procedure is executed over a grid of band widths.

Procedure	Size (alpha = 0)	Power (IR =0.75, p = 5%)
KTWW	0.66	0.80
Thresholded FF, c = 1.0	1.00	1.00
Thresholded FF, c = 2.0	0.99	1.00
Thresholded FF, c = 3.0	0.29	0.66
Thresholded FF, c = 4.0	0.09	0.40
Thresholded FF, c = 5.0	0.07	0.27
Unmodified FF (c = ∞)	0.00	0.03



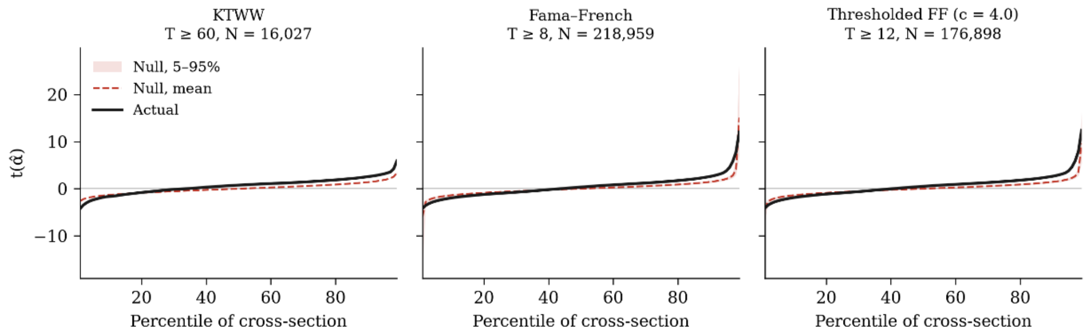
The results reproduce the diagnosis of Harvey and Liu (2022) on independent data, and in a more extreme form. Thresholding at c = 4.0 keep the nominal size and raises power relative to unmodified FF.

The calibrated value c = 4.0 exceeds the value of 2.0 that Harvey and Liu (2022) obtain for a monthly mutual fund panel data. This is consistent with their explicit warning that no bootstrap is universally optimal. The daily wallet panel is substantially sparser and more unbalanced, so a wider band is required.

There are two limitations of the calibration method. First, no single band width simultaneously calibrates all order statistics. Inference is therefore based on the 95th percentile, designated in advance. Second, the 99th percentile statistics reject every replication under every configuration. Because short-history wallets remain in the actual cross-section, but these are dropped by the >= 8 observation screening, so the simulated cross-section is with thinner tails.

6.3 Full-Sample Results

Test	N	Statistic	Actual	Null mean	p
KTWW	16,027	90th pct	2.570	1.404	<0.001
		95th pct	3.152	1.869	<0.001
FF	218,959	90th pct	2.607	1.473	<0.001
		95th pct	3.793	2.101	<0.001
HL (c = 4.0)	176,898	90th pct	2.645	1.466	<0.001
		95th pct	3.822	2.062	<0.001



The right tail of the cross-section lies far outside what luck can generate. At the 95th percentile the actual $t(\hat{\alpha})$ is 3.82 against a null mean of 2.06 under the primary specification, and no simulated cross-section in 1000 iterations attains the actual value. An FDR decomposition applied to the cross-section of two-sided p-values estimates the null proportion at $\hat{\pi}_0 = 0.505$ and recovers a proportion of wallets with genuinely positive alpha of 15.4%; 36,578 wallets are individually significant at the 5% level with a positive t-statistic.

6.4 Fees

The tests are repeated on gross returns, with the fee schedule set to 0. Fee effects are negligible. At the 95th percentile the full-sample actual statistics move from 3.822 to 3.827, and the estimated proportion of skilled wallets from 15.4% to 15.5%. It is important to notice fees applied only to a subset of market types for a small part of the sample period.

6.5 Results by Behavioral Type

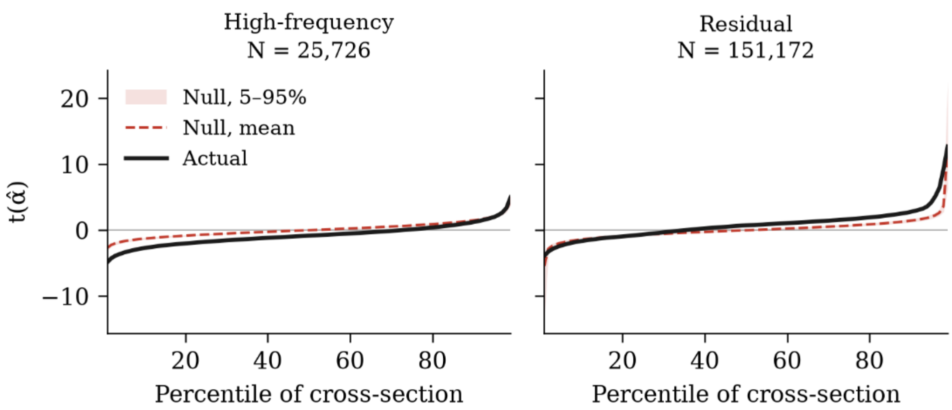
Wallets are partitioned into a high-frequency group (67,232 wallets) and a residual group (586,678). The residual group can be viewed as directional traders, high-frequency group can be viewed as high-frequency arbitrageurs and market makers. 51.1% of high-frequency wallets reach the FF sample against 31.5% of the residual group.

Statistic	High-frequency	Directional

	Actual	Null	p(L)	p(R)	Actual	Null	p(L)	p(R)
10th pct	−2.698	−1.262	0.000	1.000	−1.631	−1.385	0.045	0.955
Median	−0.813	0.021	0.000	1.000	0.751	0.009	1.000	0.000
90th pct	1.246	1.514	0.002	0.998	2.791	1.457	1.000	0.000
95th pct	2.020	2.114	0.290	0.710	4.210	2.050	1.000	0.000
Fraction $t > 0$	0.279	0.508	0.000	1.000	0.652	0.504	1.000	0.000
FDR π^{+}	3.8%				17.6%			

As stated in Section 5.5, p-values in this table are computed as k/B. The correction $(k+1)/(B+1)$ shifts every p-value by at most $1/(B+1) \approx 0.001$ and leaves all inferences unchanged.

Thresholded FF (c = 4.0), by behavioural type



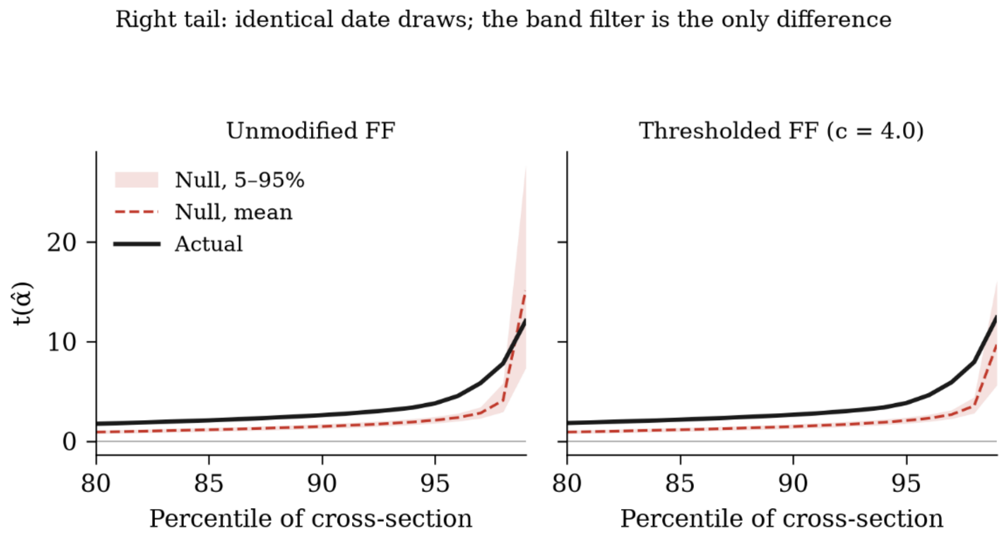
The residual group carries the full-sample result. Its distribution is shifted to the right throughout. The FDR decomposition attributes genuinely positive alpha to 17.6% of these wallets. This is skill that is moderate in magnitude and widely distributed rather than concentrated in a few specialists.

The high-frequency group displays the opposite pattern. Its distribution is shifted to the left, and the left tail lies well outside the null. The FDR decomposition attributes genuinely positive alpha to only 3.8% of the group.

Repeating the exercise on gross returns leaves the pattern intact. Removing fees entirely raises the high-frequency median from -0.813 to -0.769 and the positive fraction from 27.9% to 29.1%.

6.6 Robustness

Every result above is reproduced at a band width of c = 2.0, with similar results.



6.7 Interpretation

There are 3 findings emerging.

First, skill exists in this market, roughly one wallet in six earns genuinely positive alpha.

Second, skill is concentrated among directional traders, even though they trade in smaller size and at lower frequency than the algorithmic wallets. If this finding is robust, it overturns the prior with which we began. Directional traders remain willing to express private ability in a fully transparent venue. One reading is that the cryptocurrency segment of Polymarket remains relatively uncrowded. This also suggests that Polymarket's price-discovery role extends beyond the political-event markets.

Third, skill is not where the microstructure literature would place it. The wallets that trade most intensively lose systematically and lose by margins that luck cannot explain. A possible explanation is the maker rebate. Polymarket's Liquidity Rewards Program compensates market makers off-chain, and these payments leave no trace in the OrderFilled record.

7 Conclusion

This paper asks whether participants in Polymarket's cryptocurrency betting earn returns that luck alone cannot explain. Minute-level profit and loss are reconstructed for all 653,910 wallets from 218.9 million order-book fills, each wallet is treated as an actively managed portfolio with its BTC exposure removed, and the cross-section of risk-adjusted performance is tested against three bootstrap procedures whose size and power are calibrated on the panel itself.

Three findings emerge. Skill exists in this market: the right tail of the cross-section exceeds what luck can produce under every procedure and every specification. Skill is not, however, distributed as microstructure theory would predict. The most active participants, whose behaviour is consistent with market making and arbitrage, lose systematically and by margins that luck cannot explain, while moderate positive alpha is widely dispersed among less frequent, directionally oriented traders.

The paper also makes a methodological contribution. The three procedures examined here were developed for monthly panels of a few thousand funds; the panel studied here is daily, two orders of magnitude larger in the cross-section, and far more unbalanced. On it, the diagnosis of Harvey and Liu is reproduced in a more pronounced form: the Kosowski et al. procedure rejects in two-thirds of replications in which no wallet possesses skill, while the unmodified Fama–French procedure detects genuine skill in three per cent. Calibration selects a band width of 4.0 rather than the 2.0 obtained on monthly fund data, consistent with Harvey and Liu's own warning that no bootstrap is universally optimal.

Several limitations bound these conclusions. Firstly, A Polymarket wallet resembles a hedge fund more than a mutual fund, with a payoff profile that has non-linear exposures. The option-based factors and the timing terms of Treynor-Mazuy may be introduced in the future. Secondly, this paper establishes that skill exists, not that it persists, and it does so in a period of low taker fees and off-chain maker rebates. Polymarket has since experienced a sharp expansion in participation and several important changes in its fee schedule. Future research could treat the fee change as a regression discontinuity in time.

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Appendices

Appendix 1 Code Link

https://github.com/sethlu-lusiyu/research-project-2026

Appendix 2 Filtering Criteria

This dataset includes only markets and events tagged with "tag_id": 21 (label: Crypto), while excluding any tagged with "id": 102368 (Pre-Market) or "id": 833 (ETF).

Only markets and events that have a valid start date and resolved on or before February 5, 2026 are included. Results are further filtered by timeframe tags, defined as TAG_LABELS = {"102892": "5M", "102467": "15M", "102175": "1H", "102531": "4H", "102281": "Daily", "102264": "Weekly", "102144": "Monthly", "102536": "Yearly"}, and any markets or events associated with the target tags TARGET_TAGS = ["MicroStrategy", "Tech", "Stocks"] are removed from the final set.

Appendix 3 Data Structure from Goldsky's polymarket.order_filled

Term	Type	Meaning
id	string	Unique event identifier
block_number	long	Block number of the event
block_timestamp	long	Unix timestamp of the block
transaction_hash	string	Transaction hash
address	string	Contract address that emitted the event
user_id	string	Address of the user whose order was filled
asset	string	Token ID of the outcome token
amount_usdc	double	USDC value of the fill
amount_shares	double	Number of outcome token shares filled
price	double	Fill price (between 0 and 1)
tx_type	string	Transaction type (e.g. TRADE)
side	string	Order side (BUY or SELL)
order_hash	string	Hash of the order
counterparty_id	string	Address of the counterparty
order_type	string	Whether this order was maker or taker
fee	double	Fee paid for this fill
builder	string	Builder address if applicable
_gs_op	string	Goldsky insert & delete identifier